

Alexandre Garcia-Duran

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RESEARCH SUMMARY

PhD researcher in **computational neuroscience** at the Centre de Recerca Matemàtica (CRM, Barcelona). My research combines **probabilistic graphical models**, **machine learning**, and **dynamical systems theory** to study how the brain performs perceptual inference — particularly through the lens of bistable perception. I am also interested in adversarial robustness and connectome-constrained neural network models, which I explored during a recent visit to the **Turaga lab at Janelia Research Campus (HHMI)**.

EDUCATION

MSc Advanced Mathematics and Mathematical Engineering

2022 – 2023

Universitat Politècnica de Catalunya (UPC)

- Thesis: "An integrative model for perceptual decision-making and movement" — grade 10/10 (Honors).
- GPA 9.1 / 10.

BEng Biomedical Engineering

2018 – 2022

Universitat de Barcelona (UB)

- Thesis: "Disentangling human decisions under strong time pressure in an expectation-based 2AFC auditory task" — grade 9.7/10.
- Honors in 4 subjects: Internship program, Biomedical Signals & Instrumentation, Biological Systems Modelling, and Biomechanics.
- GPA 8.2 / 10.

RESEARCH EXPERIENCE

PhD Researcher · Centre de Recerca Matemàtica (CRM)

Nov 2023 – Present

Computational Neuroscience Group — supervised by A. Hyafil & M. Molano-Mazón

- Developing **probabilistic graphical models** of bistable visual perception and **loopy belief propagation**.
- Studying how **lateral connections** in the visual hierarchy encode statistical structure for perceptual inference.
- Presented at major conferences: COSYNE, CCN, ICMNS (5 posters + 1 invited talk in 2024–2026).

Short-term Research Visitor · HHMI Janelia Research Campus

Mar – Apr 2026

Turaga Lab

- Investigated **adversarial attacks** on optical-flow networks constrained by the *Drosophila* **connectome**.
- Explored computational links between fly visual illusions and network robustness.

Research Assistant · Centre de Recerca Matemàtica (CRM)

Jul 2022 – Oct 2023

Computational Neuroscience Group — supervised by A. Hyafil, M. Molano-Mazón & G. Huguet

- Led behavioral analysis for rat and human sensorimotor data; implemented **drift-diffusion models**.
- Co-authored a paper published in **Nature Communications** (2024).

Research Intern · IDIBAPS — Brain Circuits & Behavior Lab

Jan – Jun 2022

Group of Jaime de la Rocha

- Conducted psychophysics experiments and analyzed human behavioral data (history biases, GLMs).

Research Intern · Institute for Bioengineering of Catalonia (IBEC)

Jun – Aug 2021

Integrative Cell & Tissue Dynamics — supervised by N. R. Chahare

- Worked on mechanobiology and microfluidics experiments.

PUBLICATIONS

Molano-Mazón*, M., **Garcia-Duran*, A.**, Pastor-Ciurana*, J., et al. (2024). Rapid, systematic updating of movement by accumulated decision evidence. *Nature Communications*, 15, 10583. [DOI](#) (* equal contribution)

SELECTED POSTERS & TALKS

Talks

- "Bistable perception reveals a canonical computation in sensory processing." BARCCSYN, Barcelona, May 2026.
- "An attempt to understand (bistable) perception." SIJIMAT seminar, CRM, Barcelona, June 2025.
- "Bistable perception emerges from loopy inference in strongly coupled probabilistic graphs." Neuroscience Seminars at UPC, December 2024.

Selected Posters

- COSYNE 2025 (Lisbon, Mar 2025) — "Lateral connections in visual hierarchy encode statistical structure to support perceptual inference."
- CCN 2025 (Amsterdam, Aug 2025) — "Bistable perception emerges from loopy inference in strongly coupled probabilistic graphs."
- COSYNE 2024 (Lisbon, Feb 2024) — "Rapid, systematic updating of movement by accumulated decision evidence."
- IBRO World Congress of Neuroscience 2023 (Granada, Sep 2023) — "Fast and tight control of motor vigor by accumulated decision evidence."

FELLOWSHIPS & FUNDING

PhD Fellowship · Spanish Ministry of Science and Innovation

Nov 2023 – Nov 2027

"Deciphering the brain mechanisms for perceptual inference through bistable perception."

TECHNICAL SKILLS

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| <ul style="list-style-type: none">• Python (primary — NumPy, PyTorch, scikit-learn)• MATLAB · R• Bayesian / probabilistic inference• Variational autoencoders & Boltzmann Machines• Adversarial attacks & robustness analysis• Connectome-constrained neural networks | <ul style="list-style-type: none">• Probabilistic graphical models• Drift-diffusion & dynamical systems modelling• Generalized linear models• Psychophysics experimental design• Optical flow & image processing• Bifurcation analysis & stochastic dynamics |
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TRAINING & CERTIFICATES

- NeuroAI Summer School 2025, Champalimaud Foundation, Lisbon (July 2025).
- Barcelona Summer School for Advanced Modelling of Behaviour (BAMB!) 2024.
- Probabilistic Graphical Models Specialization — Coursera / Stanford Online (2024).
- Deep Learning Specialization — Coursera / DeepLearning.AI (2021).

COMMUNITY & SERVICE

- Co-organizer of Neurochats seminars (CRM, Barcelona) — interdisciplinary forum for PhD students and postdocs.
- Co-organizer of BARCCSYN 2025 annual meeting (Barcelona, May 2025).